

Qwen3

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Note: The Raspberry Pi 5 1GB version cannot run this case. Please refer to online large model cases. Due to performance limitations, the Raspberry Pi 5 2GB and 4GB versions need to run the small parameter version.

Qwen 3 is the latest generation of large-scale language models in the Qwen series, providing a complete suite of dense and mixture-of-experts (MoE) models.

1. Model Size

Model	Size
qwen3:0.6b	523MB
qwen3:1.7b	1.4GB
qwen3:4b	2.6GB
qwen3:8b	5.2GB

2. Performance

	Qwen3-235B-A22B <i>MoE</i>	Qwen3-32B <i>Dense</i>	OpenAI-o1 <i>2024-12-17</i>	Deepseek-R1	Grok 3 Beta <i>Think</i>	Gemini2.5-Pro	OpenAI-o3-mini <i>Medium</i>
ArenaHard	95.6	93.8	92.1	93.2	-	96.4	89.0
AIME'24	85.7	81.4	74.3	79.8	83.9	92.0	79.6
AIME'25	81.5	72.9	79.2	70.0	77.3	86.7	74.8
LiveCodeBench <i>v5, 2024.10-2025.02</i>	70.7	65.7	63.9	64.3	70.6	70.4	66.3
CodeForces <i>Elo Rating</i>	2056	1977	1891	2029	-	2001	2036
Aider <i>Pass@2</i>	61.8	50.2	61.7	56.9	53.3	72.9	53.8
LiveBench <i>2024-11-25</i>	77.1	74.9	75.7	71.6	-	82.4	70.0
BFCL <i>v3</i>	70.8	70.3	67.8	56.9	-	62.9	64.6
MultilF <i>8 Languages</i>	71.9	73.0	48.8	67.7	-	77.8	48.4

1. AIME 24/25: We sample 64 times for each query and report the average of the accuracy. AIME'25 consists of Part I and Part II, with a total of 30 questions.
2. Aider: We didn't activate the think mode of Qwen3 to balance efficiency and effectiveness.
3. BFCL: The Qwen3 models are evaluated using the FC format, while the baseline models are assessed using the highest scores obtained from either the FC or prompt formats.

	Qwen3-30B-A3B MoE	QwQ-32B	Qwen3-4B Dense	Qwen2.5-72B-Instruct	Gemma3-27B-IT	DeepSeek-V3	GPT-4o 2024-11-20
ArenaHard	91.0	89.5	76.6	81.2	86.8	85.5	85.3
AIME'24	80.4	79.5	73.8	18.9	32.6	39.2	11.1
AIME'25	70.9	69.5	65.6	15.0	24.0	28.8	7.6
LiveCodeBench v5, 2024.10-2025.02	62.6	62.7	54.2	30.7	26.9	33.1	32.7
CodeForces Elo Rating	1974	1982	1671	859	1063	1134	864
GPQA	65.8	65.6	55.9	49.0	42.4	59.1	46.0
LiveBench 2024-11-25	74.3	72.0	63.6	51.4	49.2	60.5	52.2
BFCL v3	69.1	66.4	65.9	63.4	59.1	57.6	72.5
MultitF 8 Languages	72.2	68.3	66.3	65.3	69.8	55.6	65.6

1. AIME 24/25: We sample 64 times for each query and report the average of the accuracy. AIME'25 consists of Part I and Part II, with a total of 30 questions.
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3. Using Qwen3

3.1. Running Qwen3

Use the run command to start running the model. If you haven't downloaded the model, the model from the Ollama model library will be automatically downloaded:

```
ollama run qwen3:8b
```

```

$ ollama run qwen3:8b
pulling manifest
pulling a3de86cd1c13: 100% 5.2 GB
pulling ae370d884f10: 100% 1.7 KB
pulling d18a5cc71b84: 100% 11 KB
pulling cff3f395ef37: 100% 120 B
pulling 05a61d37b084: 100% 487 B
verifying sha256 digest
writing manifest
success
>>> Send a message (/? for help)

```

3.2. Starting a Conversation

```
Please tell me how many hours there are in a day
```

Response time depends on your hardware configuration, so please be patient!

```
>>> Please tell me how many hours there are in a day
Thinking...
Okay, the user is asking how many hours there are in a day. Let me start
by recalling the basic units of time. A day is typically considered to be
24 hours. But wait, I should make sure I'm not missing any nuances here.
For example, in some contexts, like in astronomy, a day can refer to a
solar day, which is the time it takes for the Sun to return to the same
position in the sky, which is slightly longer than a sidereal day. But the
standard answer people usually give is 24 hours.

Let me think if there are any exceptions or different ways to measure a
day. There's also the concept of a day in different time zones, but that
doesn't change the number of hours in a day itself. Each time zone just
shifts the starting point of the day. So regardless of where you are, a
day still has 24 hours.

Wait, could there be a scenario where a day has more or fewer hours? For
instance, in some countries, daylight saving time might change the clock
by an hour, but that doesn't alter the actual number of hours in a day. It
just shifts the time. So even with daylight saving, a day still has 24
hours.

Another angle: some people might confuse a day with a week or a month, but
the question specifically asks about hours. So I should stick to the
```

3.3. Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

3.4. Chinese Dialogue

If you don't have a Chinese input method, please refer to the [Chinese input method switching tutorial](#).

Chinese Dialogue:

```
>>> 一个小时是多少分钟？请用中文告诉我
Thinking...
好的，用户问“一个小时是多少分钟？请用中文告诉我”。首先，我需要确认用户的问题
是否简单。一个小时等于60分钟，这是基本的常识，所以回答应该直接明了。

接下来，我需要考虑用户的背景。用户可能是在学习时间单位，或者在日常生活中需要
转换时间。比如，学生可能在做数学题，或者家长在帮助孩子做作业。也有可能用户是
外国人，正在学习中文，所以需要用中文回答，同时保持语言简洁。

然后，检查是否有其他可能的意图。用户可能只是想确认基本的时间单位转换，或者有
更深层次的需求，比如如何计算更长的时间段，但根据问题本身，应该先回答基础问题
。

另外，用户之前问过一天有多少小时，现在问一个小时有多少分钟，这可能是在学习时
间单位的转换，所以需要确保回答准确，并且可能补充一些相关的信息，比如分钟和秒
的关系，但根据问题，可能不需要扩展太多。

还要注意用户是否可能有拼写错误或误解，但问题中的“小时”和“分钟”是正确的术语，
所以不需要纠正。同时，保持回答的口语化，避免使用复杂术语，让回答更易理解。

最后，确保回答符合用户的要求，用中文，并且简洁明了。不需要添加额外信息，除非
用户有进一步的问题。所以，直接回答“一个小时等于60分钟”即可，同时保持友好和帮
助的态度。
```

References

Official website: <https://ollama.com/>

GitHub: <https://github.com/ollama/ollama>

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GitHub: <https://github.com/QwenLM/Qwen3>

Ollama corresponding model: <https://ollama.com/library/qwen3>